

P/G Theory — Economics Research Agenda

For each candidate paper: the topic in detail, the existing literature it engages, and how *dynamic* P/G could improve on it

Working memo prepared for MindOrigin (P/G Theory / P/G Valuation / Minimal P/G Affected)

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1. How to read this document

This is a forward-looking companion to the cross-disciplinary positioning memo. That memo asked *where P/G sits* in existing economics. This one asks the opposite: *what new economics papers could P/G seed*, and for each, what the existing literature is and exactly how the **dynamic** form of P/G would improve on it.

One honest constraint governs the whole document, carried over unchanged: in economics, expressiveness is a *cost* (it weakens identification and falsification), so almost every topic below routes through the **experimental, structural-estimation, or computational** wings rather than core analytical theory, and **identification** — separating the property-weight W_P from the relationship-weight W_G — is the recurring gate. Each topic is therefore tagged with a realistic venue ceiling and its key hurdle. Only two of the fourteen are plausible top-five *theory* shots; the rest are field-journal, experimental, or computational contributions. Claiming more would repeat the overreach the positioning memo warns against.

1.1 The four dynamic levers of P/G

Every “how P/G improves it” claim below draws on one or more of four properties that the existing models drop for tractability. They are referenced throughout as the *dynamic levers*.

1. **Direction.** $g_{ij} \neq g_{ji}$: A can trust B while B distrusts A. Most relational variables in economics are symmetric or scalar.
2. **Multi-dimensionality.** Trust, obligation, threat, and cooperation potential are *separate* fields, not one number; an action can move them in opposite directions at once.
3. **Decay and repair (the core dynamic lever).** A retention factor ρ governs how a relational field forms, decays, and is repaired over time. P/G models the *time-path* of a relationship, not only its current level.
4. **Threshold-gated selection.** The current property state P , evaluated against thresholds whose positions are set by personality, switches *which* relationship field G is active; the active G then drives the action. The same P can activate different G across agents — the mechanism behind behavioral diversity.

A fifth, non-dynamic property — **explicitness** (P/G is kept separate from the scalar objective and from hard constraints, so it stays inspectable) — matters for the mechanism-design and reputation topics.

2. Macroeconomics and growth

2.1 Trust as a directed, decaying micro-state: an agent-based account of the trust-growth link

The paper. Build an agent-based growth/development model in which trust is not a scalar national average but a population of directed, decaying edges g_{ij} . Let agents form, exploit, and break trust through interaction; aggregate the edge population into a measured “generalized trust” index; and test whether the endogenous dynamics reproduce the observed trust-TFP relationship — then use the model to identify *which* channel (firm-internal trust, market trust, institutional trust) carries the aggregate effect.

Existing related work.

- Knack & Keefer (QJE 1997) — generalized trust correlates with growth.
- Algan & Cahuc (AER 2010; *Ann. Rev. Econ.* 2013) — causal trust→growth via inherited trust; effect on TFP and firm organization.
- Tabellini (JEEA 2010); Guiso, Sapienza, Zingales (JEEA 2016) — culture / civic capital and outcomes.
- Bloom, Sadun, Van Reenen — trust and firm decentralization (the firm-level channel).

How dynamic P/G improves it. The literature observes trust only in aggregate and as a level (lever 3 absent, lever 1 absent). It cannot say whether the trust-growth link runs through asymmetric principal-agent trust, symmetric peer trust, or institutional trust, because the scalar erases direction and channel. A directed, decaying edge population (levers 1 + 3) lets the *same* aggregate index decompose into channels, and lets the model generate the time-path — trust build-up, betrayal, slow repair — that a static regressor cannot. The deliverable economics cannot currently produce is a *mechanism* whose aggregate output matches the reduced-form trust-growth fact.

Ceiling & hurdle. *Ceiling:* JEDC / JEBO / computational economics; top-five only if a novel aggregate prediction is validated on data. *Hurdle:* disciplined aggregation and a genuine data match, not merely plausible dynamics.

2.2 Persistence of mistrust as retention-regime heterogeneity

The paper. Reframe the documented persistence of mistrust as heterogeneity in the retention factor ρ across relationship contexts, and ask what retention structure reproduces observed half-lives of mistrust after a historical shock.

Existing related work.

- Nunn & Wantchekon (AER 2011) — the slave trade and the persistence of mistrust.
- Voigtländer & Voth; Alesina, Giuliano (JEL 2015) — persistence of cultural attitudes.
- Bisin & Verdier (JET 2001) — cultural transmission dynamics.

How dynamic P/G improves it. This literature establishes *that* mistrust persists but

models persistence as a fixed coefficient on a historical instrument; it does not mechanize the decay (lever 3). P/G's ρ is exactly a per-context retention mechanism: high- ρ , slow-decay edges in violated/adversarial contexts versus low- ρ edges elsewhere. The improvement is a *runnable* decay-and-transmission model whose parameters can be fit to the observed half-life, rather than a single persistence coefficient.

Ceiling & hurdle. *Ceiling:* economic-history / field journal. *Hurdle:* identifying ρ from historical series.

Boundary. Analytical growth theory proper (Solow, Romer, Aghion-Howitt) is *not* a target: P/G adds nothing to a balanced-growth path, factor shares, or the ideas-production function.

3. Repeated games, reputation, and theory

3.1 Relationship-context differentiation as an equilibrium-selection criterion

The paper. The folk theorem leaves a continuum of equilibria. Ask whether requiring agents to carry multi-relationship state and to differentiate behavior across relationship contexts — formalized as the context-differentiation quantity $D_{c,c'}$ — selects among them, i.e., whether “stabilize *differently* across contexts” is a refinement that picks a unique or smaller equilibrium set.

Existing related work.

- Fudenberg & Maskin (Econometrica 1986) — the folk theorem; the multiplicity to be resolved.
- Kreps, Milgrom, Roberts, Wilson (JET 1982) — reputation via commitment types.
- Abreu, Pearce, Stacchetti (Econometrica 1990) — recursive methods / self-generating sets.
- Mailath & Samuelson (2006) — reputations synthesis.

How dynamic P/G improves it. Standard refinements (renegotiation-proofness, Markov perfection) are context-blind. P/G's differentiation condition is a *relationship-typed* selection device: among payoff-equivalent equilibria, prefer the one whose policy differentiates appropriately across friend/stranger/adversary contexts (levers 2 + 4). If this can be shown to select, it is a genuine theory contribution — not a re-description.

Ceiling & hurdle. *Ceiling:* a top theory journal **if** a clean selection theorem is proved; GEB if only a representation result. *Hurdle:* it must yield a theorem, not a restatement. Highest ceiling, highest risk on the list.

4. Network economics

4.1 Vector-valued directed relational networks: per-dimension centrality and intervention

The paper. Extend the key-player / centrality framework to networks whose edges are vector-valued (trust, obligation, threat) and decaying, and show that the optimal intervention — whom to target to shift aggregate behavior — differs from the scalar-edge prediction when the dimensions interact.

Existing related work.

- Ballester, Calvó-Armengol, Zenou (Econometrica 2006) — equilibrium action $\propto$ Bonacich centrality; the “key player.”
- Bramoullé, Kranton, D’Amours (AER 2014) — strategic substitutes/complements on networks.
- Galeotti, Golub, Goyal — optimal network interventions.
- Acemoglu, Carvalho, Ozdaglar, Tahbaz-Salehi (Econometrica 2012) — micro network structure $\rightarrow$ macro fluctuations (the aggregation bridge).

How dynamic P/G improves it. These models carry a *single* scalar edge weight and one interaction type. P/G edges are multi-dimensional, directed, and decaying (levers 1-3). The same centrality machinery applies, but to a richer object: centrality can be computed *per relationship dimension*, and the key-player can differ across dimensions (the trust-key-player is not the threat-key-player). The improvement is an intervention theory that the scalar-edge model gets wrong precisely when dimensions trade off.

Ceiling & hurdle. *Ceiling*: top field journal; top-five if the intervention result is clean and new. *Hurdle*: it must produce a result the scalar model gets wrong, not merely a richer representation.

5. Behavioral and social-choice economics (the most realistic cluster)

5.1 A structural P/G social-preference model that beats Fehr-Schmidt out-of-sample

The paper. Specify a structural utility over (P, G) states — directed, multi-dimensional, with a decay term — estimate its weights on repeated-trust-game and gift-exchange data, and show it predicts held-out behavior (trust repair, conditional cooperation, betrayal response) better than inequity-aversion.

Existing related work.

- Fehr & Schmidt (QJE 1999) — inequity aversion (two scalar parameters).
- Bolton & Ockenfels (AER 2000); Charness & Rabin (QJE 2002) — richer social-preference functions.
- Berg, Dickhaut, McCabe (GEB 1995) — the trust game (estimation data).
- Fehr, Kirchsteiger, Riedl (QJE 1993) — gift exchange.

How dynamic P/G improves it. Existing social-preference models are scalar and static: inequity aversion has no direction, no separation of trust from obligation, and no decay/repair path (levers 1–3 all absent). P/G’s directed, multi-dimensional, decaying preference structure is richer on exactly the axes that govern *repeated* interaction — which is where Fehr–Schmidt is weakest. The improvement is measured as out-of-sample predictive accuracy on repeated games, the cleanest possible economics test.

Ceiling & hurdle. *Ceiling:* AER/QJE behavioral, *Experimental Economics*, GEB — genuinely top-tier-reachable. *Hurdle: identification* — you need designed variation that moves P and G independently to separate W_P from W_G ; adding dimensions makes this harder, not easier. This is the single best “real economics paper” P/G can seed.

5.2 Commitment versus sympathy, made estimable

The paper. Operationalize Sen’s distinction — sympathy (others’ welfare in your utility) versus commitment (counterpreferential choice) — as, respectively, an other-P weight and a G-dominant decision, and design an experiment that separates them behaviorally.

Existing related work.

- Sen, “Rational Fools” (*Phil. & Public Affairs* 1977) — the conceptual distinction, never operationalized.
- Andreoni (warm-glow giving); Bénabou & Tirole (AER 2006) — prosocial motivation.
- Akerlof & Kranton (QJE 2000) — identity in the utility function.

How dynamic P/G improves it. Sen diagnosed that a single preference ordering is overloaded but gave no computable structure. P/G separates self-P, other-P, and shared-G as distinct weighted channels (lever 2), with commitment representable as a G-dominant decision where shared-G outweighs self-P. The improvement is turning a philosophical critique into an estimable model with a testable prediction.

Ceiling & hurdle. *Ceiling:* behavioral / experimental field journal. *Hurdle:* designing choices that cleanly separate commitment from sympathy.

5.3 Threshold-gated relational mode switching (the distinctive topic)

The paper. Test the P/G decision mechanism directly as a behavioral hypothesis: humans do not respond to a situation with a fixed policy; the current property state P , crossing a threshold whose location is set by stable personality traits, switches *which* relationship frame is active, and the active frame drives behavior. Design a paradigm where the *same* externally-presented P state (e.g., a fixed level of threat or resource scarcity) produces systematically different relational behavior across personality-measured subjects, and where manipulating P across a threshold flips behavior within subject.

Existing related work.

- Kahneman & Tversky (Econometrica 1979) — reference dependence and thresholds in valuation.
- Construal-level / appraisal theory (psychology) — framing as a function of state.
- Behavioral game theory on type heterogeneity (level- k , social-value-orientation measures).
- Your own P/G Valuation §4 (Behavioral Diversity Principle) and the Minimal P/G Affective Core (personality and thresholds as layers above the valence core).

How dynamic P/G improves it. This is the one topic that is *not* a re-derivation of an existing economics model — it tests lever 4, threshold-gated frame selection, which no standard social-preference or game-theoretic model contains. Existing models explain heterogeneity by *different weights*; P/G explains it by *the same agent selecting a different frame at a threshold set by personality*. The improvement is a mechanism for behavioral diversity that makes a within-subject, threshold-crossing prediction the weights-heterogeneity account cannot make.

Ceiling & hurdle. *Ceiling*: behavioral / psychology-and-economics. *Hurdle*: measuring the threshold and the trait *independently* of the behavior they are meant to predict. This is the most distinctive paper on the list and the one I would prioritize alongside 5.1.

6. Experimental methodology

6.1 A relational-impact elicitation protocol

The paper. A methods contribution: instruments and a coding schema that record not a single trust score but *which* G dimension moved, *for whom*, *by how much*, and *over what horizon* — plus a public dataset and inter-rater reliability analysis.

Existing related work.

- Berg, Dickhaut, McCabe (GEB 1995) — the trust game (single dependent variable).
- Fehr & Gächter (AER 2000; *Nature* 2002) — cooperation and punishment paradigms.
- The annotation/agreement methods your own P/G Theory §5.3 already specifies (Cohen's κ , Krippendorff's α).

How dynamic P/G improves it. Existing paradigms collapse the relational outcome to one number, discarding direction, dimension, and horizon (levers 1–3). A multi-dimensional, directed, time-stamped dependent-variable schema is the measurement substrate every other topic in this document needs. The improvement is resolution: it makes the structural estimation (5.1) and the mechanism test (5.3) possible.

Ceiling & hurdle. *Ceiling*: *Experimental Economics* / methods. *Hurdle*: inter-rater reliability on subjective relational dimensions. Lowest-risk topic on the list; it unlocks the others.

7. Family economics

7.1 Multi-dimensional altruism

The paper. Decompose the single altruism weight of the Becker tradition into *whose* P, *which* dimension (safety vs. autonomy vs. reputation), and *which* relationship type, and test whether parents trade off a child's safety against the child's autonomy in ways a scalar-altruism model cannot represent.

Existing related work.

- Becker, "A Theory of Social Interactions" (JPE 1974); the Rotten Kid Theorem.
- Bergstrom — Rotten Kid analyses.
- Cox; Altonji, Hayashi, Kotlikoff — empirical intergenerational transfers.

How dynamic P/G improves it. Becker's altruism is one weight on another's *scalar* utility. P/G makes the other enter multi-dimensionally and dynamically (levers 2 + 3): the kin tie has its own state and can erode after betrayal, and the altruist can weight a child's safety differently from the child's autonomy. The improvement is the ability to represent and test trade-offs *within* the altruistic relationship.

Ceiling & hurdle. *Ceiling:* household-economics field journal / experimental. *Hurdle:* identification and data.

8. Information economics and signaling

8.1 Joint-effect signaling

The paper. Model and lab-test signals that move *multiple* relational dimensions at once — a display that deters (raises perceived threat) while simultaneously damaging trust — and characterize when a sender uses such a signal despite the trust cost.

Existing related work.

- Spence, "Job Market Signaling" (QJE 1973) — costly signaling (single signaled quality).
- Crawford & Sobel (Econometrica 1982) — cheap talk.
- Zahavi (handicap principle) — the biology twin already cited in the papers.

How dynamic P/G improves it. Signaling models compress to one signaled quality. P/G represents what the signal changes — displayed P and perceived G — as explicit, separable, accumulating state (levers 2 + 3), so a single act can raise threat while lowering trust. The improvement is a model of joint-effect signals that single-quality models structurally cannot carry.

Ceiling & hurdle. *Ceiling:* theory or experimental field journal. *Hurdle:* a clean equilibrium characterization or a clean experiment.

9. Development economics

9.1 Informal insurance with decaying obligation edges

The paper. Model village risk-sharing as a network of obligation-G edges with retention ρ , and predict the *repair* dynamics of the insurance network after a shock — who re-enters which obligations, how fast — testing against panel data.

Existing related work.

- Townsend, “Risk and Insurance in Village India” (Econometrica 1994).
- Fafchamps & Lund — risk-sharing networks.
- Banerjee, Chandrasekhar, Duflo, Jackson (Science 2013; REStud 2019) — diffusion and gossip over real networks.

How dynamic P/G improves it. This literature models risk-sharing as an equilibrium or estimates a static network; it does not model how obligations *decay and re-form* after a shock (lever 3). P/G’s decaying obligation edges give an explicit repair-dynamics prediction. The improvement is a time-path for informal-institution recovery, not a one-shot estimate.

Ceiling & hurdle. *Ceiling:* development field journal; top-five if validated on panel data. *Hurdle:* data, and identifying decay separately from re-formation.

10. Evolutionary game theory

10.1 Content-ful conventions

The paper. Standard convention-selection theory asks *which* abstract convention is selected under noisy adaptation. Ask the selection question when conventions carry P/G *content* — what the convention is *about* (a deference norm, a trust-repair norm) — with retention dynamics, and characterize stochastic stability for content-bearing conventions.

Existing related work.

- Kandori, Mailath, Rob (Econometrica 1993); Young (Econometrica 1993) — stochastically stable equilibria / evolution of conventions.
- Maynard Smith; Nowak — the biology twins already cited.

How dynamic P/G improves it. Convention-selection theory treats conventions as abstract labels. P/G supplies their content and a retention mechanism (levers 2 + 3), so the framework can ask which *kind* of relational convention is selected and how fast it decays. The improvement is content-and-decay where the existing theory has labels.

Ceiling & hurdle. *Ceiling:* theory field journal. *Hurdle:* a clean stability result.

11. Mechanism design and industrial organization (AI-economics adjacent)

11.1 Designing auditable relational objectives for artificial agents

The paper. Treat the design of an artificial agent’s objective as a mechanism-design problem: feasibility constraints (the safe set A_{safe}) plus a *structured* relational objective E_{PG} whose components are inspectable, and ask when such an objective is incentive-compatible and non-gameable.

Existing related work.

- Myerson, Maskin, Hurwicz (mechanism design; Nobel 2007); the revelation principle.
- Reward-hacking / Goodhart and the Lucas critique (the failure mode).

How dynamic P/G improves it. Standard objectives are scalar and opaque; the designer cannot see *which* relational dimension an action is rewarded for. P/G’s explicit, multi-dimensional objective (lever 2 + explicitness) is auditable by construction — the designer can inspect and constrain per dimension. The improvement is an objective whose components a designer can verify, which matters for artificial agents in a way it does not for human institutions.

Ceiling & hurdle. *Ceiling:* EC (Economics & Computation) / AI-economics venues, not core economics. *Hurdle:* a real incentive-compatibility result.

11.2 Multi-dimensional reputation systems

The paper. Replace the scalar star-rating of platform reputation systems with a directed, multi-dimensional standing vector (reliability, honesty, responsiveness), and analyze its incentive and welfare properties versus the scalar benchmark.

Existing related work.

- Tadelis (*Ann. Rev. Econ.*) — reputation and online marketplaces.
- Dellarocas — digital feedback mechanisms.

How dynamic P/G improves it. Star ratings collapse multi-dimensional standing into one number and are typically symmetric (levers 1 + 2 absent). A directed, multi-dimensional, decaying reputation vector is closer to how trust actually behaves and is directly useful in the interactive-agent products where P/G is strongest. The improvement is a reputation object that supports per-dimension incentives.

Ceiling & hurdle. *Ceiling:* IO / market-design field journal. *Hurdle:* showing the multi-dimensional system improves incentives rather than just adding complexity.

12. Prioritization and honest tiering

Start here (lowest risk, genuinely publishable, mutually reinforcing):

1. §6.1 elicitation protocol — builds the measurement substrate the others need.

2. **§5.1 structural P/G social preferences vs. Fehr-Schmidt out-of-sample** — the best shot at a top-tier behavioral paper.
3. **§5.3 threshold-gated mode switching** — the most *distinctive* topic, because it tests the P/G mechanism rather than re-deriving an existing model.

High ceiling, high risk (the only plausible top-five *theory* shots): §3.1 (equilibrium selection by context-differentiation) and §4.1 (per-dimension centrality and intervention) — each lands only if it produces a genuine theorem/result, not a representation.

Computational / field-journal (niche but real): §2.1, §2.2, §9.1, §10.1.

AI-economics adjacent (EC / AAMAS, not core economics): §11.1, §11.2.

Off-limits: analytical macro/growth core — P/G adds nothing there, and targeting it would invite immediate rejection.

The throughline, unchanged from the positioning memo: nearly every topic is contingent on running an experiment and on earning identification, and the genuine P/G-distinctive contributions are the threshold-gated-switching test (§5.3) and the directed, multi-dimensional, decaying structural estimation (§5.1). *Resolution is a cost in economics and a benefit in AI* — so even the successful versions of these papers are contributions to the experimental, computational, and behavioral wings, and the framework's larger payoff remains on the AI-agent side.